Course: Communication Theory

Assignment 1

Task : You are required to simulate and compare two quantization techniques: a uniform quantizer and a machine-learning-based quantizer using K-means clustering.

Create m-file named with your first and last name

- Generate a random input vector of 10000 samples randomly in the range from [-a, a] where a is the largest number of the group members IDs i.e. if the two IDs are 31-2380 and 31-5694→ a = 9
- Simulate your quantizer three times with number of levels that are equal to 2, 4 and 8

The m-file output should include:

1. The quantized vectors obtained using both the **uniform quantizer** and the **K-means-based quantizer** for L=2,4 and 8, respectively. For each case, display only the first six elements of the original input vector and the corresponding quantized vectors.

2. Calculate the **mean absolute quantization error** for both the uniform and K-means-based quantizers for L=2,4 and 8. Plot the mean absolute quantization error versus the number of levels for **both quantizers on the same figure** and comment on their performance.

3. Calculate the **variance of the quantization error** for both quantizers for L=2,4 and 8. Plot the obtained variance versus the number of levels for both quantizers. For the **uniform quantizer**, also plot the theoretical variance of the quantization error on the same figure. Comment on the obtained results.

4. Calculate the **SQNR** for both the uniform and K-means-based quantizers for L=2,4 and 8. Plot the SQNR versus the number of levels for both quantizers. For the

uniform quantizer, also plot the theoretical SQNR on the same figure. Compare the results and comment on the performance of both techniques.

- Based on the obtained results, briefly explain the difference between the two quantization approaches and how the quantization levels are determined in each case.

Deliverables

1. The m-file sent to email: sama.ashraf@guc.edu.eg

2. A printed report including the m-file, all the output figures, the values and the comments stated above

Notes

- The code written should be adaptable to any number of levels given to you i.e. you should write one code and change only the number of levels to get the output.
- **Any report written by any AI tool will take zero.**
- **Any copied assignments will take zero.**
- **Any line of code written should be fully understood by the students; otherwise, marks will be deducted during the evaluation**.

Submission

- Submission will be in the office C4.329 (back)
- The assignment will be completed in groups of up to **three students. The same groups will remain unchanged for the following project and for the rest of the semester.**
- **Deadline: 9 November 2026. Any report submitted after the deadline will not be accepted.**